

BE Minor Project Proposal Defence

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS**



**Domain-Adaptive Deep Learning for Massive
MIMO OFDM Channel Estimation Using Measured
CSI**

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1. Introduction

- Massive MIMO + OFDM form the core air interface of 5G and are key technologies for future 6G networks.
- Uplink pilot signals are used to estimate downlink CSI through channel reciprocity, significantly reducing pilot overhead.
- Traditional CSI estimation techniques such as Least Squares (LS) and Minimum Mean Square Error (MMSE) suffer performance degradation

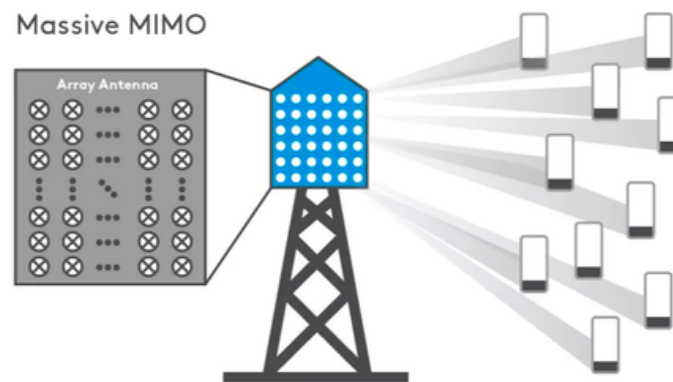


Figure1 : Massive MIMO antenna array layout.

2. Problem Statement



1. Deep-learning CSI estimators are trained almost entirely on synthetic data.
2. This causes poor generalization to real, measured CSI due to the domain gap.
3. Most deep models are black boxes : they don't exploit known channel physics (sparsity, spatial correlation).
4. Supervised adaptation to real data would need labeled real CSI, which is expensive/impractical to collect at scale.



3. Objectives

General Objective:

The main objective of this project is to develop a channel-aware, domain-adaptive deep learning framework for Massive MIMO–OFDM channel state information (CSI) estimation, leveraging inherent wireless channel properties to enable robust generalization from simulated to real measured environments.

Specific Objective:

1. To understand the characteristics and challenges of channel estimation in Massive MIMO–OFDM systems.
2. To explore deep learning techniques for improving domain generalization in channel estimation across diverse wireless channel environments.
3. To analyze the effectiveness of the developed framework using simulated and real-world wireless channel data

4. Literature Review Summary

Method / Paper	Main Contribution	Limitation
LS / MMSE	Classical channel estimation methods. LS is computationally simple, while MMSE provides improved estimation accuracy.	LS is noise-sensitive. MMSE has high computational complexity.
ChannelNet (CNN)	Uses cascaded CNNs to refine coarse LS estimates into accurate CSI reconstruction.	Requires large labeled datasets and has limited generalization.
Joint DL Beamforming	Performs channel estimation and hybrid beamforming simultaneously using deep learning.	Performance depends on training data quality.
LDAMP	Combines sparse recovery with CNN-based denoising for beamspace channel estimation.	Assumes sparse channels and requires iterative processing.
GRU / LSTM	Models temporal variations in time-varying wireless channels.	Higher training and inference complexity.
Attention Models	Enhances feature extraction by focusing on important channel information.	Additional computational overhead.
Deep-mMIMO-WNOMA	Employs wavelet-domain representation for improved noise robustness.	Higher implementation complexity than FFT-based methods.
DeepMIMO	Provides ray-tracing-based datasets for benchmarking deep learning models.	Simulation-to-reality gap.

Table 1: Literature Review

5. Gap Analysis



Existing Work	Limitation
CNN / ResNet-based CSI Estimation	Primarily trained and evaluated on synthetic datasets; limited generalization to real measured channels.
MMSE and Statistical Estimators	Require accurate channel statistics; computationally expensive in large-scale Massive MIMO systems.
Deep Learning Channel Estimation	Often operates as a black box; does not explicitly exploit channel sparsity and spatial correlation.
Domain Adaptation Methods	Limited work on adaptation from simulated to real measured CSI without labeled target-domain data.

Table 2: Gap Analysis of Existing Works

6. Research Design and Workflow

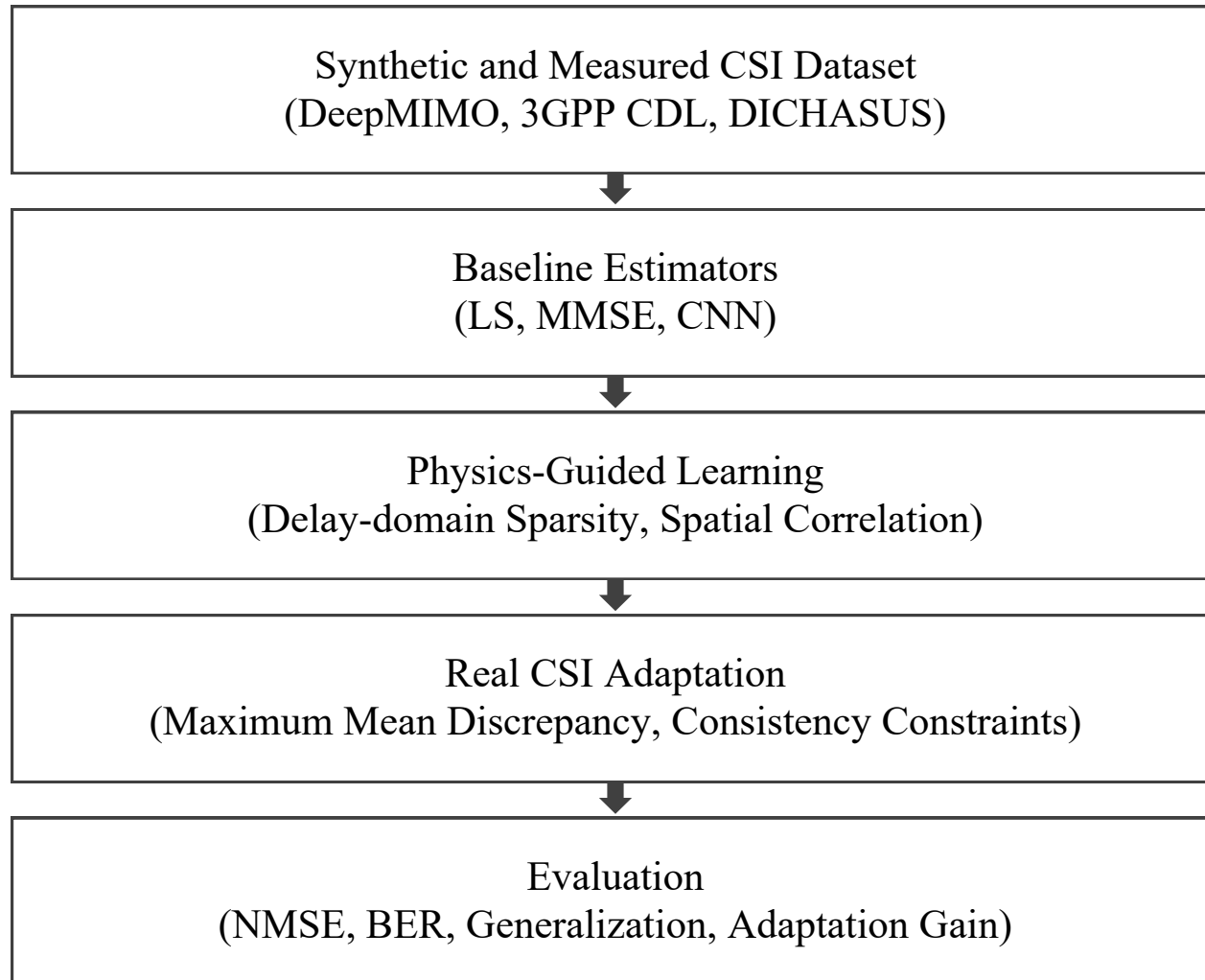


Figure 2: Multi-stage framework for physics-guided domain-adaptive CSI estimation



7. Proposed Methodology

Stage 1 - Dataset Preparation:

Generate synthetic Massive MIMO-OFDM CSI (DeepMIMO, 3GPP CDL) and preprocess real CSI (DICHASUS).

Stage 2 - Baseline Channel Estimation:

Compute LS/MMSE estimates and train a lightweight CNN using synthetic CSI.

Stage 3 - Physics-Guided Channel Learning:

Apply delay-domain sparsity and spatial correlation constraints during CNN training.

Stage 4 - Self-Supervised Domain Adaptation:

Adapt the pretrained CNN to real CSI using MMD-based feature alignment and consistency learning.

Stage 5 - Performance Evaluation:

Evaluate channel estimation performance using NMSE, BER, adaptation gain, and generalization metrics.

8. System Architecture

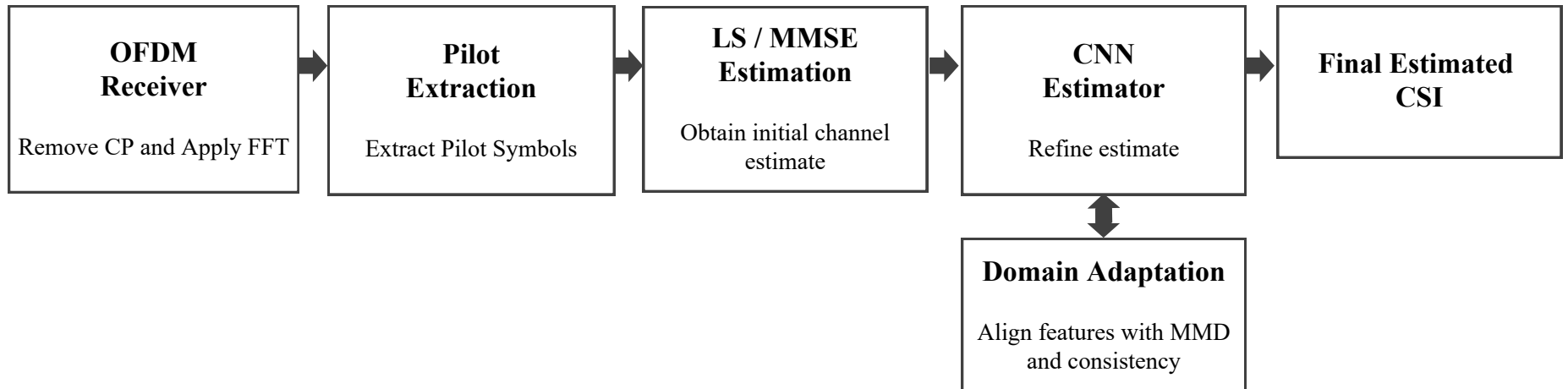


Figure 3: System architecture of the proposed CSI estimation framework

9. Algorithms and Models



- **CNN**

- DL model based on convolutional layers.
- Specialized for structured grid-like data.
- Captures patterns across antennas and subcarriers.

- **ResNet**

- Enhanced CNN with residual (skip) connections.
- Supports deeper network training.
- Provides more accurate channel estimates in challenging conditions.

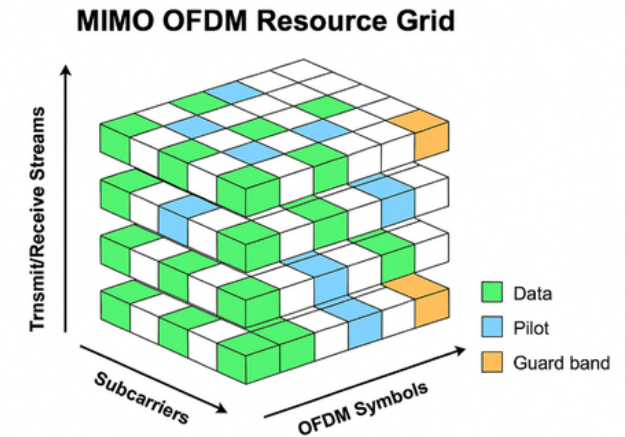


Figure 4: OFDM resource grid and Channel estimation



10. Physics-Guided Loss Function

$$\mathbf{L}_{\text{Total}} = \mathbf{L}_{\text{MSE}} + \lambda_{\text{Sparse}} \mathbf{L}_{\text{Sparse}} + \lambda_{\text{Corr}} \mathbf{L}_{\text{Corr}}$$

$\mathbf{L}_{\text{MSE}}$	Minimizes error between estimated and true CSI.
$\mathbf{L}_{\text{Sparse}}$	Enforces delay-domain sparsity using an L1 penalty.
$\mathbf{L}_{\text{Corr}}$	Maintains spatial correlation between antenna elements.

Delay-Domain Sparsity

- Channel energy is concentrated in a few dominant delay paths.
- Used to guide the model toward physically realistic channel estimates.

10. Physics-Guided Loss Function

Spatial Correlation

- Neighboring antennas exhibit similar channel characteristics.
- Used to preserve the spatial structure of Massive MIMO channels.

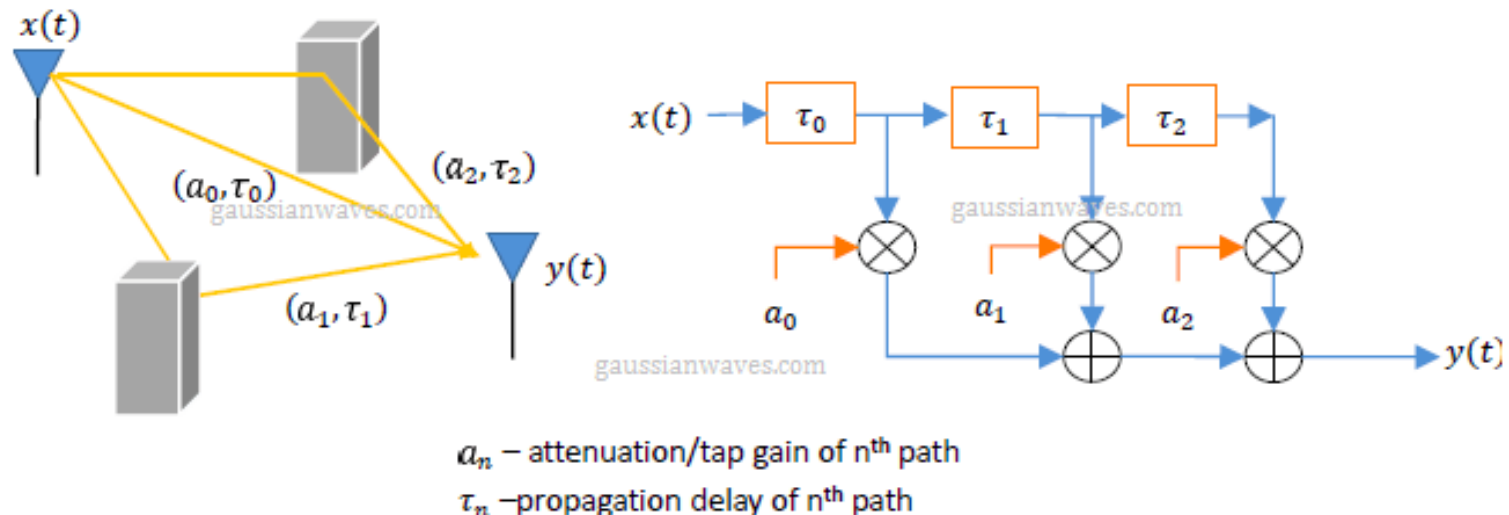


Figure 5: 3-ray multipath time-invariant channel and its equivalent TDL implementation (path attenuations a_n and propagation delays τ_n are fixed)

11. Self-Supervised Domain Adaptation

$$L_{Total} = L_{MSE} + \lambda_{Sparse} L_{Sparse} + \lambda_{Corr} L_{Corr} + \lambda_{MMD} L_{MMD} + \lambda_{Consistency} L_{Consistency}$$

Maximum Mean Discrepancy (MMD)

- Compares synthetic and real feature distributions.
- Reduces the gap between both domains.
- Improves adaptation to measured CSI.

Consistency Regularization

- Adds noise or masks pilot symbols.
- Forces similar predictions for both inputs.
- Improves robustness to input variations.

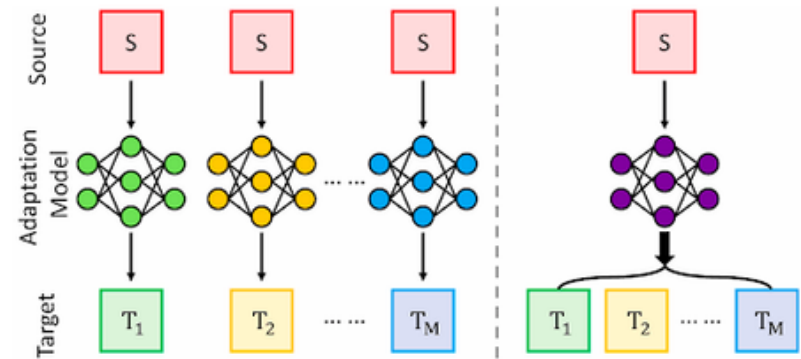


Figure 6: Domain Adaptation Model

12. Dataset and Data Acquisition Plan



- **Synthetic Datasets**

- DeepMIMO: a widely used ray-tracing dataset for massive MIMO channels.
- 3GPP CDL: clustered delay line models for 5G NR evaluation.

- **Real CSI Dataset**

- DICHASUS (DICHASUS-005x): over-the-air measured Massive MIMO-OFDM channel responses.
- Collected using a 32-antenna array with 1024 subcarriers; used as the target domain for adaptation.



13. Time Schedule

Week	Activities
Week 1	Literature review; dataset collection (DeepMIMO, 3GPP CDL, real CSI dataset).
Week 2	Implementation of LS and MMSE estimators; initial data preprocessing.
Week 3	Development of CNN baseline model and training on synthetic data.
Week 4	Design and integration of physics-guided loss functions.
Week 5	Implementation of self-supervised adaptation on real CSI data.
Week 6	Comprehensive experiments (NMSE, BER, generalization), report writing, and presentation preparation.



14. Expected Outcomes

1. Unified CSI estimation framework
2. Domain-adaptive learning approach
3. Physics-aware deep learning model
4. Improved estimation performance
5. Comprehensive experimental evaluation



15. Feasibility

1. Uses mature tools: DeepMIMO, 3GPP CDL simulators, PyTorch/TensorFlow
2. Benchmarked against proven CSI estimators: LS, MMSE, and CNN
3. Utilizes the publicly available DICHASUS over-the-air CSI dataset
4. Structured 6-week development timeline (Data Preparation → Baseline Models → Physics-Guided Learning → Domain Adaptation → Evaluation)



16. Conclusion

- **Deep learning-based CSI estimation** : offers high accuracy but suffers from the synthetic-to-real domain gap and limited physical interpretability.
- **Proposed framework:** Combines physics-guided learning with domain adaptation using self-supervised MMD.
- **Expected improvements:** Lower NMSE and BER, with better generalization
- **Project impact:** Bridges the gap between simulation-based CSI estimation and real-world wireless deployment

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THANK YOU

Questions and Suggestions are Welcome